

SALES ANALYSIS

February 2026 · eCommerce Orders

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EDA · Data Quality · Insights · Recommendations

Structured exploratory analysis of 621 orders | 2 markets | 6 categories

Dataset Structure and Scope

• Period

Feb 1 – Mar 1, 2026 (1 month)

• Records

722 lines · 621 unique orders

• Markets

US and MX (+ 4 minor countries)

• Channels

US, MX, External US, External

• Categories

Smoked Salt, Bundle, Seasoning, Sauce, Snack, Accessory

• Payments

Standard, CreditCard, GiftCertificate, Installments

20 AVAILABLE COLUMNS

order_id · order_date · updated_date

status · channel · fulfillment

shipping_tier · product_id · category

quantity · currency · unit_price

tax · shipping_cost · discount

city · state · country

payment_method · is_business

Issues Found in the Dataset

Field	Records affected	% of total	Note
tax	626 / 722	86.7 %	Not applicable to all channels/markets
shipping_cost	511 / 722	70.8 %	Likely absorbed by platform fulfillment
discount	718 / 722	99.4 %	Only 4 lines; fixed \$71.28 discount (one-off promotion)
currency / unit_price	57 / 722	7.9 %	No sale price → excluded from the revenue calculation
quantity = 0	51 / 722	7.1 %	Cancelled or adjustment lines — see assumptions
city / country	3–4 / 722	0.4 %	Records without complete geolocation

⚠ The dataset is not perfect, but it is interpretable. The issues are consistent with real-world e-commerce data.

Assumptions and Analysis Decisions

01 MXN → USD exchange rate

A rate of \$20 MXN per USD was used to standardize revenue across markets. MXN figures were converted only for comparison; the per-market analysis keeps the original currency.

03 Volume outliers (External channel)

2 orders have quantities of 112 and 51 units (External channel). They were identified as likely B2B/wholesale sales and documented, but not excluded from the order count.

02 Records with quantity = 0

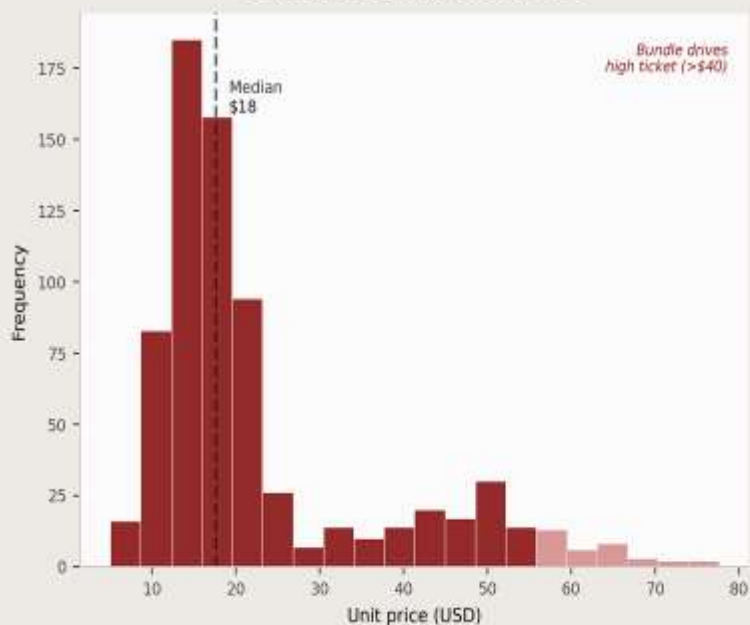
51 lines have zero quantity but a 'Shipped' status. They were interpreted as cancelled lines or inventory adjustments and excluded from the revenue calculation.

04 Cost fields not used in revenue

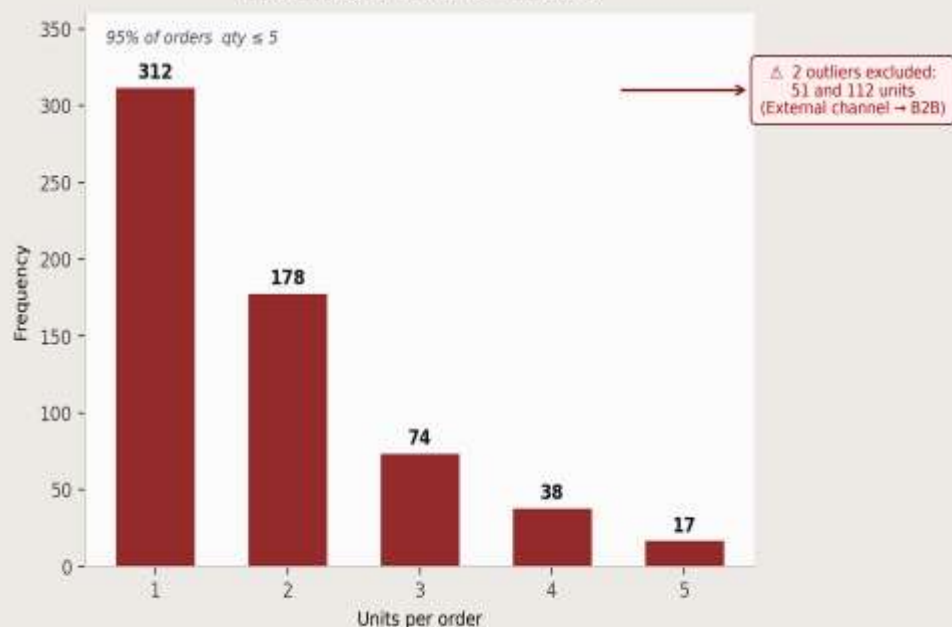
Tax and shipping_cost are >70% null. Gross revenue was calculated solely as $\text{unit_price} \times \text{quantity}$ to keep consistency across the whole dataset.

Distributions and Outliers

Unit Price Distribution (USD)



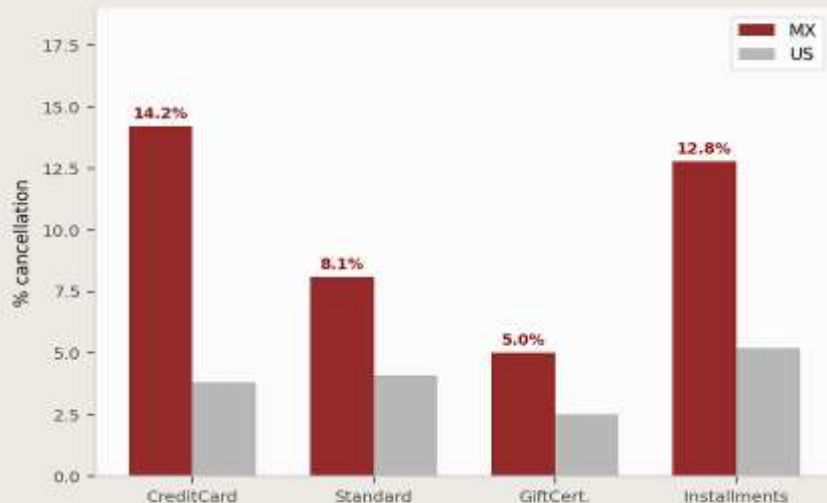
Quantity per Order Distribution (excluding wholesale outliers)



💡 Unit price concentrated in \$10–\$20 (long tail toward Bundles >\$40). Quantity per order: 95% ≤ 6 units — 2 B2B outliers with 51 and 112 units in the External channel.

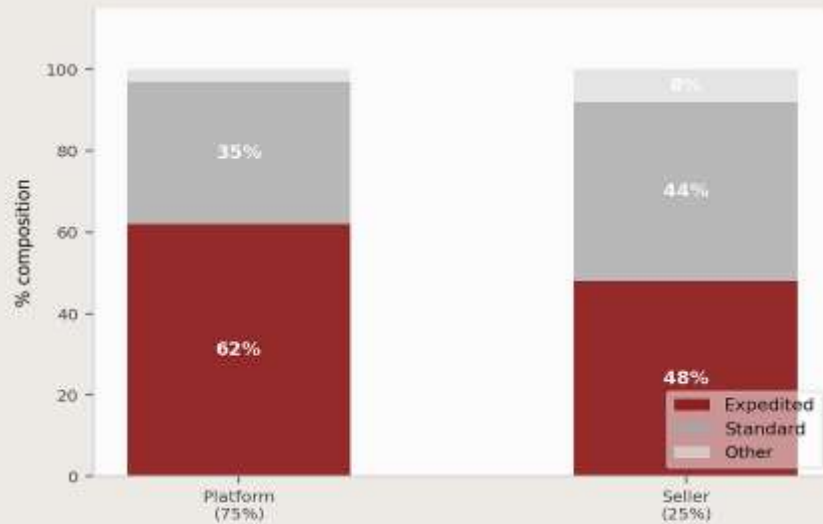
Relationships Between Variables

Cancellation Rate by Payment Method (%)



▲ CreditCard and Installments carry the highest cancellation rate in MX — possible friction in the payment flow

Fulfillment vs Shipping Type (% order composition)



Platform uses more Expedited (62%) → higher shipping cost absorbed by the channel; Seller has more Standard (44%)

💡 CreditCard and Installments carry the highest cancellation rate in MX (14.2% and 12.8%). Platform fulfillment uses more Expedited (62%) vs Seller (48%).

Business Overview · February 2026

621

Unique orders

722 lines in total

~\$16.6K

Estimated revenue (USD)

Excludes qty=0 and no price

2

Main markets

US and MX nearly even

6

Categories

Bundle leads in \$

7.1%

Lines with qty=0

Likely cancellations/adjustments

10.9%

Cancellation rate in MX

vs 4.4% in the US

75%

Platform fulfillment

vs 25% Seller

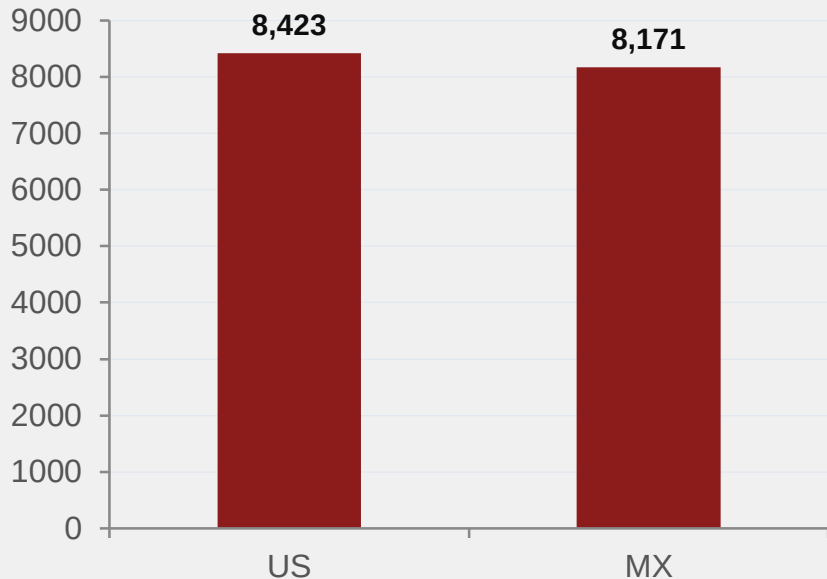
59%

Expedited shipping

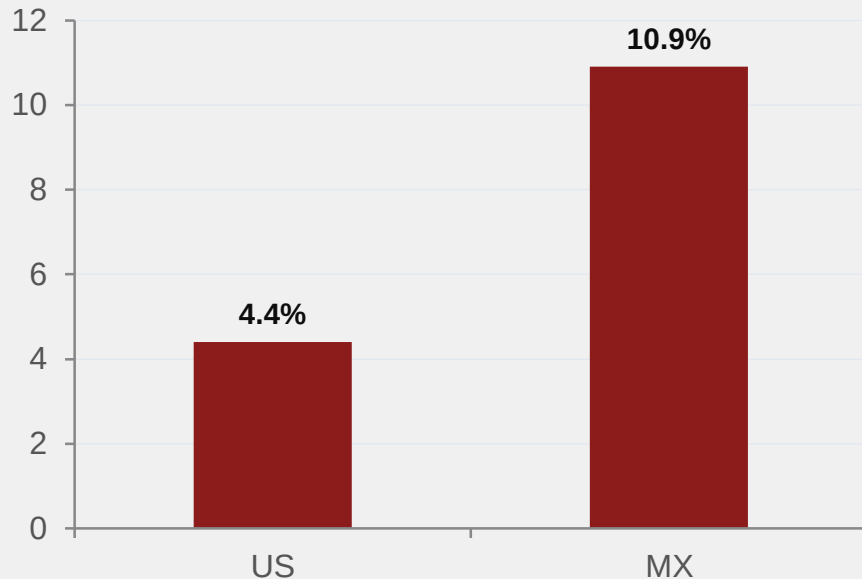
vs 39% Standard

Market Comparison: US vs MX

Estimated revenue (USD)



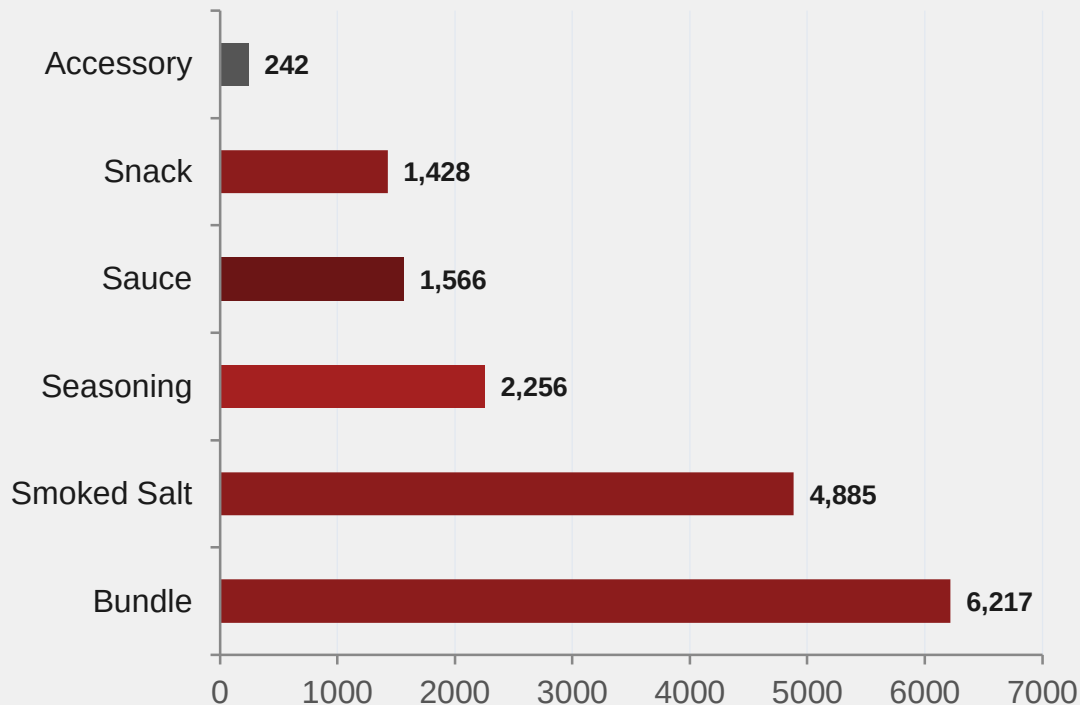
Cancellation rate (%)



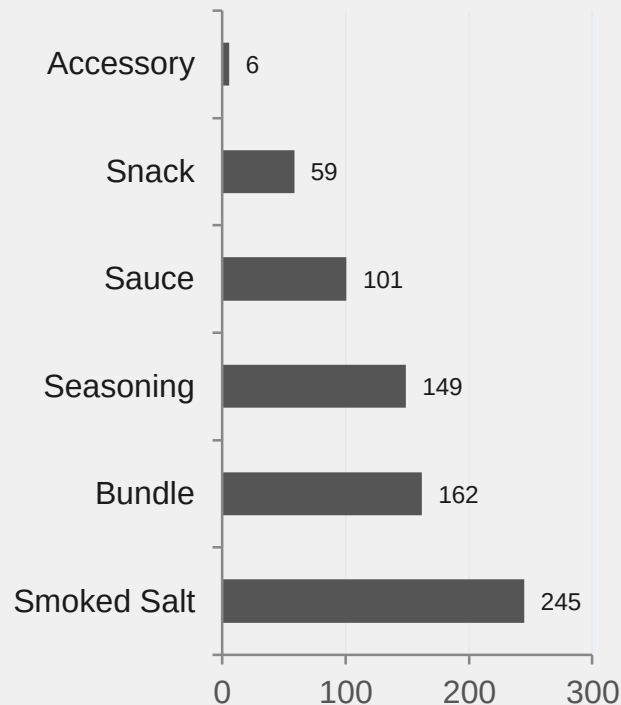
💡 Similar revenue across markets (~\$8.4K vs \$8.2K USD), but MX has 2.5× more cancellations than the US (10.9% vs 4.4%). Causes to investigate: logistics, payment experience, or product.

Performance by Product Category

Estimated revenue by category (USD)

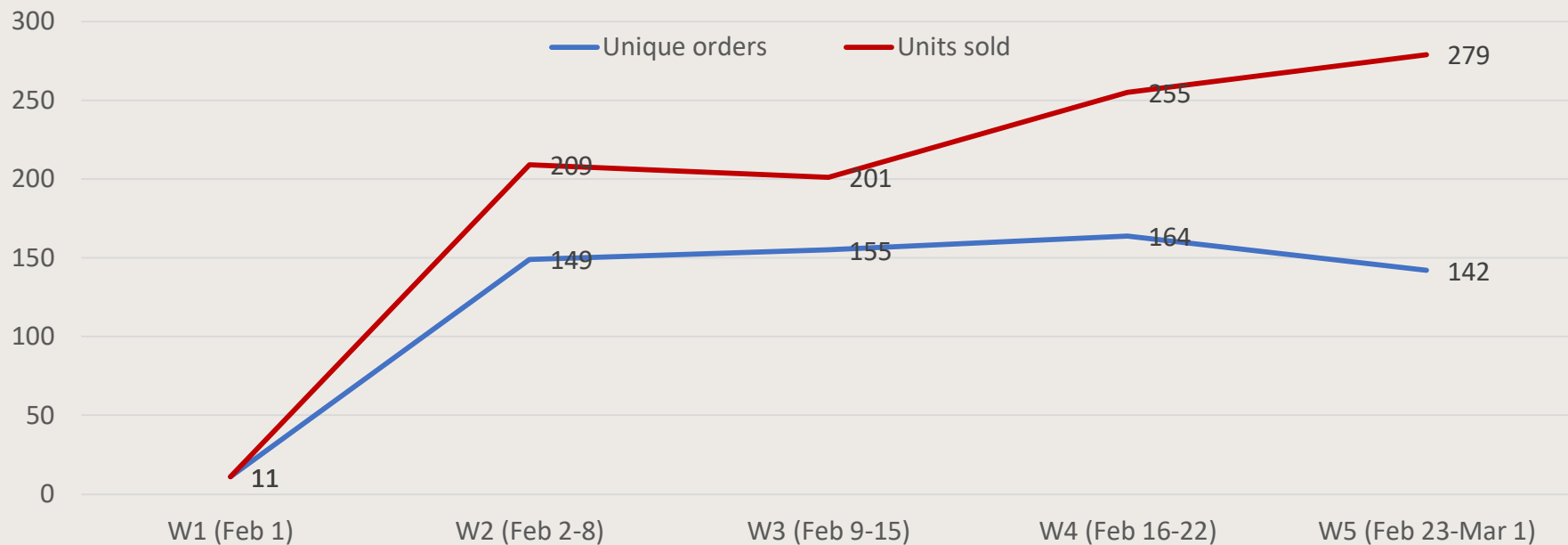


Volume (# lines)



💡 *Bundle generates the highest revenue (\$6.2K) but has fewer lines than Smoked Salt (245 lines). Bundle has the highest average ticket in the portfolio.*

Weekly Order Trend



*W1 contains only February 1 (partial week) · W5 includes March 1

W4 peak in units (255)
W5 upward trend in items vs orders → average basket rising

C0 LAB Analisis_Ventas_Feb2026.ipynb

Python • pandas • matplotlib • seaborn

```

import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# General style
sns.set_theme(style="whitegrid", palette="muted")
plt.rcParams['figure.dpi'] = 100
plt.rcParams['font.size'] = 11

# Load and initial diagnosis
df = pd.read_csv('ecommerce_orders_feb2026.csv')
print(f"Lines: {df.shape[0]} | Cols: {df.shape[1]}")
df.head(5)

# Quality: nulls and key unique values
nulls = df.isnull().sum()
pct = (nulls / len(df) * 100).round(1)
for col in ['status', 'channel', 'fulfillment']:
    print(df[col].value_counts())

# Quantity outliers (wholesale / B2B)
df[df['quantity'] > 10][['order_id', 'quantity', 'channel']]

```

```

# Cleaning: exclude qty=0 and no price
df_clean = df[(df['quantity'] > 0) &
              df['unit_price'].notna()].copy()

# Gross revenue + MXN - USD conversion ($20)
df_clean['revenue'] = (df_clean['unit_price'] * df_clean['quantity'])
df_clean['revenue_usd'] = df_clean.apply(
    lambda r: r['revenue'] / 20 if r['currency'] == 'MXN'
    else r['revenue'], axis=1)

# Market comparison: US vs MX
mx_us = df_clean[df_clean['channel'].isin(['US', 'MX'])]
ingreso = mx_us.groupby('channel')['revenue_usd'].sum()
cancel = df[df['channel'].isin(['US', 'MX'])]
c_rate = cancel.groupby(['channel', 'payment_method'])
        ['status'].apply(lambda x: (x == 'Cancelled').sum() / len(x) * 100)

# Performance by category + avg ticket
cat_stats = df_clean.groupby('category').agg(
    ingreso_usd=('revenue_usd', 'sum'),
    lineas=('order_id', 'count'), ticket=('revenue_usd', 'mean'))
.sort_values('ingreso_usd', ascending=False)

# Weekly trend (ISO week: Feb 2026 -> weeks 5-9)
df_clean['week'] = df_clean['order_date'].dt.isocalendar().week.astype(int)
weekly = df_clean.groupby('week').agg(
    ordenes=('order_id', 'nunique'), unidades=('quantity', 'sum'))

```

Suggested actions from the analysis

1 Reduce cancellations in MX

The 10.9% rate in MX is roughly 2.5× that of the US. Review the payment flow (heavy local CreditCard use), delivery-zone coverage, and communication of delivery times.

2 Strengthen Bundle as a revenue anchor

Bundle generates 37% of revenue with only 22% of the lines. Cross-sell strategies with Smoked Salt (higher volume) could raise the average ticket.

3 Investigate lines with qty = 0

51 lines with zero quantity and a 'Shipped' status are an anomaly. They may represent data-entry errors, late cancellations, or inventory adjustments — operational validation required.

4 Improve data capture

The tax, shipping_cost, and discount fields are up to 99% null. Define a minimum data standard so that real margin and shipping ROI by channel can be calculated.